ناطق · NAATIQ — DOCUMENTATION

Product Requirements Document (v2)

Naatiq (ناطق) · updated to reflect what was actually built and shipped

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1. Summary

Naatiq is a WhatsApp agent that turns a spoken conversation into a professional, bilingual (Arabic + English) CV — with no app to install, no forms to fill, and no literacy or typing barrier.

A user sends a WhatsApp voice note in their own Arabic dialect describing what they do for a living. The agent interviews them conversationally, extracts their skills and experience, generates a polished PDF CV, and delivers it back over WhatsApp. An impact dashboard, a public landing page and an employer portal sit on top of the same data.

> **Change from v1:** the product was originally named *Wathefti* (*وظيفتي*). It was renamed to **Naatiq (ناطق)** — "articulate" / "speaking" — which better captures the voice-first premise. The rename is applied throughout: the agent persona, the CV footer, the dashboard and all docs.

2. Goals

Shipped

- Accept an Arabic voice note over WhatsApp and understand it. **Done.**
- Conduct a warm, conversational interview that progressively builds a structured profile. **Done.**
- Generate a professional bilingual PDF CV and deliver it over WhatsApp. **Done.**
- Provide a live dashboard visualising the impact story. **Done** — Arabic-first with an English toggle.

Added during the build (not in v1)

- **Public landing page** with live counts and a WhatsApp call to action.
- **Employer portal** — browse workers, open their CV, request contact through Naatiq.
- `placements` **table** so the employment figure is real rather than invented.
- **Self-test diagnostic endpoints** on the webhook, which proved decisive in debugging.

Deferred

- **Arabic voice replies (TTS).** Fully built and deployed, gated behind a feature flag. Blocked only by ElevenLabs requiring a paid plan for API voice access. Flip `WATHEFTI_VOICE_REPLIES=true` on a paid plan and it works with no code change.
- **Mock interview practice.** Not built.
- **Job-family matching.** Table exists; matching logic not built.

Non-goals (unchanged)

User authentication, a mobile app, a full employer ATS, payments, production-grade scaling.

3. Target user and primary journey

Primary persona: a worker with real experience but no CV — domestic worker, driver, salon technician, tradesperson. May have limited literacy and only a low-cost smartphone. Speaks Arabic dialect, not formal written Arabic.

The journey (validated end to end):

1. The user messages the Naatiq WhatsApp number.
2. The agent greets them warmly in Arabic and explains it will help build a CV.
3. The user replies with voice notes in their own dialect.
4. The agent asks one question at a time — name, trade, employers, skills, tools, a proud moment.
5. Once enough is gathered, the agent says the CV is being prepared and sends a **PDF**.
6. Employers browse the resulting profiles and request contact through Naatiq.

4. Functional requirements

Area	Requirement	Status
Messaging	Receive WhatsApp text + voice via webhook; send text and media	Shipped
Voice	Transcribe Arabic-dialect voice notes accurately	Shipped
Interview	Warm Arabic persona, one question at a time, robust to messy answers	Shipped
Interview	Per-user state machine and prompt-injection resistance	Shipped
CV	Bilingual content generation, truthful, no fabrication	Shipped
CV	PDF with correct Arabic RTL shaping and embedded fonts	Shipped
Dashboard	Read-only impact visualisation from live data	Shipped
Landing	Public page with live counts and WhatsApp CTA	Shipped

Area	Requirement	Status
Employers	Browse workers, open CVs, request contact	Shipped
Voice replies	Arabic TTS responses	Built, flag-gated
Mock interview	Practice questions with feedback	Not built
Job matching	Match profiles to GCC job families	Not built

5. Data model

- **`users`** — `whatsapp_number` (unique), `display_name`, `state`, `created_at`. State machine: `interviewing → profile_complete → cv_sent → practice_mode`.
- **`profiles`** — `full_name`, `first_name` (generated), `role_trade`, `employers` (jsonb), `skills`, `tools`, `achievement`, `raw_extracted`, `cv_pdf_path`.
- **`conversations`** — `direction`, `medium` (voice/text), `transcript`, `created_at`.
- **`matches`** — job-family scoring (reserved; matching not implemented).
- **`placements`** — `employer_name`, `job_title`, `placed_at`. Powers the honest employment count.

Public surfaces never read these tables directly. They read three **read-only views** (`dashboard_stats`, `dashboard_profiles`, `dashboard_trade_distribution`) that expose first names only and never phone numbers.

6. Success criteria

Criterion	Result
A real person can send a voice note and receive a usable bilingual PDF CV in WhatsApp	**Met**
The dashboard tells the impact story with real data	**Met**
The system runs on free tiers with no manual babysitting	**Met**
A 2-minute demo video shows the magic moment	Outstanding

7. Known risks and how they were handled

- **Arabic RTL rendering — the #1 risk in v1.** Handled by choosing Gotenberg (Chromium), which shapes Arabic correctly, and by validating the design with a rendered sample early rather than the night before submission.
- **Edge Function execution time.** The webhook acknowledges Twilio immediately and does the slow work (transcription, LLM, CV) in a background task, so the ~15s webhook timeout is never at risk.

- **Prompt injection.** All user message content is wrapped and treated strictly as data; the system prompt instructs the model never to follow instructions found there.
- **Demo-day network risk.** The dashboard embeds a data snapshot and renders instantly, then refreshes live on top — so it never shows an error on stage.